User Guide: CSV to Estimate Output

Distribution Analysis & Navigation Algorithm (D.A.N.A.) Framework

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Component:	CSV to Estimate Output Guide	**Input File:**	Shift Impact CSV

Welcome to the CSV to Estimate Output Guide. This guide provides structured steps and practical scenarios for library project managers to transition raw computational data into actionable, on-the-ground shelving estimates. Be aware that you may be required to make changes to the estimate provided by the workbook. It is recommended that smaller shifts are attempted with this framework first to better understand how this tool works, as well as how you can work with this tool. I made it and still had to adjust my thinking.

> **A NOTE ON REAL-WORLD SHIFTING & LIMITATIONS**
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> This guide provides examples based on methods and scenarios that have worked for us. However, each library shift is different and must be planned according to its unique circumstances.
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> At this point in the framework, you should have the data needed to calculate a shift. However, please keep the following realities in mind:
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> - **Variables & Unknowns:** There are simply too many real-world variables to create a single guide that fits every situation. You will likely encounter unique circumstances in your library that you must adapt to on the fly (Periodicals, growing collections & circulation).
> - **Continuous Learning:** Major shifts can take months to complete, and almost every shift teaches us something new. It is impossible to anticipate or experience every possible scenario in advance.
> - **Accumulated Surplus Space:** Even when measuring out shelf targets directly, we have noticed that our shifts seem to naturally accumulate more surplus space than planned. Measuring a range side immediately after completing a shift highlights this. Minor shelf-level variations quickly add up to a major overall deviation across an entire range.

1. Importing Data into the Target Workbook

To automatically format the downloaded `Shift_Impact_Ranges_xx_to_xx.csv` file from the Library Space Management Tool, follow these exact data placement rules:

1. Locate and open the pre-formatted workbook called “Formatted Estimate From Rawdata CSV” located in the Distribution Analysis & Navigation Algorithm Framework folder.
2. Open your downloaded `Shift_Impact_Ranges_xx_to_xx.csv` file using Excel or Google Sheets.
3. Copy all of the active data held within that CSV spreadsheet.
4. Return to the destination workbook, select the sheet tab labeled **“Rawdata”**, and paste the copied data.
5. Click on the **“Dashboard”** sheet in the **“Formatted Estimate From Rawdata CSV”** workbook to view the updated estimate guide.

> **CRITICAL DATA ALIGNMENT STEP**
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> In your formatted destination workbook, you must copy and paste the corresponding **“Average Shelf Length”** values to their correct range sides used for the shift. This data was entered in your original Data Collection Workbook column ‘H’.

2. Navigating Shifting Estimates and Buffers

The dashboard translates raw mathematical linear numbers into physical operations on the library floor.

2.1 The “Moved In” and “Moved Out” Columns

These columns highlight the precise physical areas that the D.A.N.A. program allotted to be moved in order to satisfy your shifting targets.

2.2 Movable Capacity (Shift Target)

What it means: The exact number of empty shelves available to absorb moved materials. *Note: This number will not match the total of empty shelves counted after each range.*

How to use it: This is the primary number you plan against. If this number is equal to or less than zero, there will not be enough space to continue shifting materials as planned. When a negative value appears in your ‘Movable Capacity’ column, it signals that you may need to "backshift" (move collections in reverse order). To accomplish this, adjust the functional formula in the ‘Movable Capacity’ column.

2.3 Target Available (Percentage of Empty Space Per Shelf)

What it means: The required empty space set aside for each shelf within a range (e.g., leaving 25% of each shelf empty for collection growth post-shift).

2.4 Physical Empty Shelves (Accuracy Check & Large Surplus Recalculation)

What it means: The total number of physically empty shelves you will count after shifting a range. It includes both your Movable Capacity plus the Target Available space.

How to use it: Use this number when auditing shelves per range after shifting to confirm your physical floor counts match the spreadsheet.

Basic Formula Explanation:

The basic system formula `=I3+(D3/H3)` calculates total physical empty shelves by combining two parts:

- `D3/H3` **(Target Available in.):** Converts your targeted space per shelf in inches (`D3`) into shelf units by dividing by the average shelf length (`H3`).
- `I3 + (D3/H3) = (Physical Empty Shelves)`: Adds your targeted space per shelf back to your net movable space estimate (`I3`). This figure for Physical Empty Shelves is close to what you will see if you were to count empty shelves during the middle of a shift.

> **MANUAL CALCULATION FOR LARGE SURPLUS SHIFTS**
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> **Important Notice:** If a shift generates more surplus space than can fit within a single range side, the initial figure estimated for Physical Empty Shelves will be inaccurate. When dealing with large surplus shifts, use this adjusted calculation for a more accurate post-shift count:
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> 1. Total the surplus space from the 'Initial Need/Surplus' column for each range empty after shifting material to fill your first range.
> 2. Add the 'Target Available' space from any range sides that are now left empty.
> 3. Divide this combined total by the Average Shelf Length.
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> *Example:* When shifting a range to a target of 25% full, the initial system estimate projected 57.7 empty shelves. However, the actual real-world physical count was 145.5 empty shelves. Applying the manual calculation above yielded a far more accurate estimate of 146.4 empty shelves. Unfortunately, I cannot account for how much empty space your shift will start with. I am unable to automatically set a calculation for you.
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> **Second Notice:** If you notice significant unexpected changes in your shelf count after filling a range, pay attention to the Moved In/Moved Out columns. If you are pulling materials from a couple ranges away, the figures in these two columns will impact your shelf count. This sounds obvious, but I have caught myself forgetting this when I become too focused on other projects.

3. Handling Real-World Shelf Logistics

Because it is virtually impossible during physical shifts to isolate and relocate only the exact linear space required, the guide assumes every shelf will be shifted to your chosen target percentage. It leaves it to the project lead to decide where to leave surplus space behind.

> **BUFFER RECOMMENDATION**
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> We highly recommend building in an extra **3%–5% buffer** on top of the estimated physical move targets. Any surplus space accumulated beyond this 3-5% buffer should be left behind in the immediate range currently being shifted.

> **SPECIAL CONSIDERATION: PERIODICALS**
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> Shifting periodicals requires extra care and can significantly alter your project estimates:
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> - **End-to-End Shelving:** Periodicals are often packed tightly (end-to-end), which drastically changes density compared to leaving a shelf 75% filled.
> - **Future Expansion Space:** If you have growing serials, you must explicitly account for the loss of space for future volumes. Failing to account for the extra space that periodical growth needs can quickly throw off your physical capacity estimates.